

- Metodología Bottom – Up.
- Evaluación de modos de fallos a nivel componente para determinar sus efectos a nivel local, tarjeta y Sistema (identificar puntos simples de fallo).
- Estudio exhaustivo previo de la arquitectura a analizar.
- Estudio funciones (Bloques funcionales) y su dependencia/interfaces.
- Identificación de Redundancias, Medidas de mitigación, Detectabilidad, Fases Operativas (Operational, Maintenance, Debug, etc.)
- Identificación componentes/funciones críticos.
- Outputs para análisis FTA.
- A basis for design and location of performance monitoring and fault sensing device and other built-in automatic test equipment

- An FMEA is used to support the safety assessment process by providing failure rates to quantify the basic events of the FTA
- detection methods include detection by hardware or software monitors, flight crew detection, power up tests, and maintenance checks.
- When the analysis is extended to classify each potential failure mode according to its severity and probability of occurrence, it is called a Failure Mode, Effects, and Criticality Analysis (FMECA).

Documentos

targets-by-id > 1MC3gsk350qC6LiPP0KJevq6ZiOVTIFvq > 02_INGENIERIA > 00_TEMPLATES > PLANTILLAS DOCUMENTACIÓN > FMECA_FMES > Templates

	Nombre	Fecha de modificación	Tipo	Tamaño
	old	20/06/2023 13:02	Carpeta de archivos	
	1 - FMECA_Template_v3.1_1	19/06/2023 13:24	Hoja de cálculo habilita...	2.861 KB
	2 - FMES_Template_v1.6	13/07/2020 9:59	Hoja de cálculo habilita...	101 KB
	Revisiones para el FMECA_v1	05/11/2020 16:15	Documento de Microsof...	16 KB
	Z - FMECA_v1_0 - Instrucciones	15/06/2023 10:03	Documento de Microsof...	24 KB

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Schematic Reference	Item/Component	Description	Failure Ref	Component Type	PIN	Failure Mode	Failure Cause	Operational Mode	Failure Effect on Local Component	Failure Effect on AVIO2353	Failure Effect on FLAGS System
C1	Def: capacitor 1uf P/N: 011.809-PF	Function: Power Management Schematic Sheet: 10-	10.10.1.1	Ceramic Chip Capacitor	N/A	Capacitance Variance	Ceramic Chip Capacitor CAPACITANCE VARIANCE	All	No effect	No effect	No effect
C1	Def: capacitor 1uf P/N: 011.809-PF	Function: Power Management Schematic Sheet: 10-	10.10.1.2	Ceramic Chip Capacitor	N/A	Short Circuit	Ceramic Chip Capacitor SHORT CIRCUIT	All	Shortcircuit +6,3V to GND (P1V0) [U9_INT]	Loss internal supply of FPGA	Loss of AVIO2353 operations
C1	Def: capacitor 1uf P/N: 011.809-PF	Function: Power Management Schematic Sheet: 10-	10.10.1.3	Ceramic Chip Capacitor	N/A	Open Circuit	Ceramic Chip Capacitor OPEN CIRCUIT	All	No effect	No effect	No effect

< >
FMECA
FM Effect
PartNum
BOL2
FM Ratio
Hoja1
Refs
I-O
+
:
◀

1. Schematic Reference: ID componente en BOM – Esquemático.
2. Item/Component: Referencia del fabricante para cada componente.
3. Description: Descripción del componente, Hoja de esquemático, Función (bloque funcional) asignada al componente.
4. Failure Ref: Referencia numérica única para cada modo de fallo de cada componente. El valor de esta referencia depende del número de tarjeta y número de función asignada (ver hoja “Refs”).
5. Component Type: Tipo de componente (Hoja “FM Ratio”).
6. PIN: ID de los pines en IC.
7. Failure Mode: Modos de fallo para cada tipo de componente (Hoja “FM Ratio”).
8. Failure Cause: Causa de fallo (Generalmente coincide con FM).
9. Operational Mode: Identificación del modo o fase operativa del componente. Generalmente “All” salvo en casos en los que se conozca su fase operativa (Configuration, Debug, Maintenance)
10. Failure Effect at Local level: Descripción del fallo a nivel local “L3” (componente/señal)
11. Failure Effect at Board level: Descripción del fallo a nivel bloque funcional/tarjeta “L2” (bloque/señal)
12. Failure Effect at System level: Descripción del fallo a nivel sistema “L1” (Interfaces Tarjeta/Sistema FLAGS)

- * Los campos de color “Azul claro” se autocompletan (no borrar formulas) a partir de los datos del resto de hojas (“FM Effect”, “PartNum”, “BOL2”, “FM Ratio”, “Refs”, “I/O”).
- Se deben completar los campos de color “Azul oscuro”.

(13) (14) (15) (16) (17) (18) (19) (20) (21) (22) (23) (24)

Ref - Failure Effect on Local Component	Ref - Failure Effect on AVIO2353	Ref - Failure Effect on FLAGS System	Mitigation Means	Detection	Latent Time	Failure Mode Ratio [%]	Estimated Failure Rate [1/OH]	Corrective Action	Additional Remarks	Comments	Columna
AVIO2353-L3.10.105	AVIO2353-L2.10.1	FLAGS-L1.01.1		N/A		62,78	0,033737294	N			0,021181455
AVIO2353-L3.10.177	AVIO2353-L2.10.49	FLAGS-L1.01.2		External and PBIT, CBIT		31,24	0,033737294	N			0,010538557
AVIO2353-L3.10.105	AVIO2353-L2.10.1	FLAGS-L1.01.1		N/A		5,98	0,033737294	N			0,002017281

>	<u>FMECA</u>	FM Effect	PartNum	BOL2	FM Ratio	Hoja1	Refs	I-O	+	:	◀	
---	--------------	-----------	---------	------	----------	-------	------	-----	---	---	---	--

13. Ref. Failure effect at local level “L3”: Referencia numérica única para cada efecto local
14. Ref. Failure effect at board level “L2”: Referencia numérica única para cada efecto board
15. Ref. Failure effect at system level “L1”: Referencia numérica única para cada efecto sistema
16. Mitigation Means: Medidas de mitigación implementadas en la arquitectura (redundancia, back-up, monitorización, etc)
17. Detection: Métodos de detección (IBIT, CBIT, PBIT) implementados. Datos de cliente.
18. Latent Time: Tiempo transcurrido entre la aparición del fallo y su detección . Datos cliente.
19. FM Ratio: Distribución de la tasa de fallo de un componente en sus Modos de fallo.
20. Estimated FR: Tasa de fallo de cada componente
21. Correction Action: Medidas correctoras implementadas para mitigar el fallo (Automatic/Manual/None).
22. Additional Remarks: Comentario específico para cada componente.
23. Comments: Comentario general para los efectos nivel sistema “L1”.

- * Los campos de color “Azul claro” se autocompletan (no borrar formulas) a partir de los datos del resto de hojas (“FM Effect”, “PartNum”, “BOL2”, “FM Ratio”, “Refs”, “I/O”).
- Se deben completar los campos de color “Azul oscuro”.

L3-Effect	L3 effect in FMECA	L3-Nº	L2-Effect	L2 effect in L2L1 Table	L2-Effect	L2 effect in FMECA	L2-Nº	L1-Effect	L1 effect in L1L0 Table	L1-Effect	L1-Nº	Phase	Detection	Comments	Latent
Loss of Power 3.3V Aux. No effect, loss of primary power required.	Yes	1	No effect	Yes	No effect	Yes	1	No effect	Yes	No effect	1	All	N/A		
Loss of Power 12V Aux. No effect, loss of primary power required.	Yes	2	No effect	Yes	Loss of Channel 2 ARINC429 TX ARINC429 available through channels 1-3-4-5-6-7-8.	Yes	2	Loss of ARINC429 TX Channel 2	Yes	Loss of AVID2353 operations	2	All	External and PBIT, CBIT		
Loss of Power 12V Aux. No effect, Pin additional failures 1A, 1B, 1C, 2A, 2B and 2C.	Yes	3	No effect	Yes	Loss of Channel 1 ARINC429 TX ARINC429 available through channels 2-3-4-5-6-7-8.	Yes	3	Loss of ARINC429 TX Channel 1	Yes	Loss of MFCC8558/AVID2353 Maintenance operations	3	Maintenance	N/A		
Loss of Test Point, TP1. No effect.	Yes	4	No effect	Yes	Loss of Channel 4 ARINC429 TX ARINC429 available through channels 1-2-3-5-6-7-8.	Yes	4	Loss of ARINC429 TX Channel 4	Yes	Loss or incorrect Elapse Time Counter - Temperature and Maintenance Information	4	All	N/A	The effect will depend on the software finally uploaded to the computer by the client.	
Loss of Power 12V Aux. No effect, Pin additional failures 1E, 1F, 1G, 2E, 2F and 2G.	Yes	5	No effect	Yes	Loss of Channel 3 ARINC429 TX ARINC429 available through channels 1-2-4-5-6-7-8.	Yes	5	Loss of ARINC429 TX Channel 3	Yes	Loss of ARINC429 TX Channel 2	5	Operational	PBIT, CBIT		

Hoja “FM Effect”: en esta hoja se enlazan:

- los efectos locales “L3” a un efecto tarjeta “L2”
- los efectos tarjeta “L2” a un efecto sistema “L1”

La plantilla detecta si hay efectos repetidos a cualquier nivel.

Antes de definir un nuevo efecto a cualquier nivel, se debe comprobar si ya están definidos y se pueden reutilizar. De esta forma evitamos efectos duplicados/innecesarios.

B	C	D	E	F
0 empty 2 Dupl	1 Err		0 empty	0 Err
L3-Effect	L3 effect in FMECA Dupl	L3-Nº	L2-Effect	L2 effect in L2L1 Table
No effect	Yes 2 Dupl	1	No effect	Yes
no effect	Yes 2 Dupl	2	No effect	Yes
<u>no effect.</u>	No	3	No effect	Yes
No effect (Decoupling Capacitor)	Yes	4	No effect	Yes

Hojas necesarias para la generación de la hoja "FMEA" ("BOL2", "FM Ratio", "Refs")

- Hoja "PartNum": en esta hoja se listan los P/N (ref. fabricante) únicos (eliminar duplicados).
- Se asocia a cada P/N un tipo de componente definido en la Hoja "FM Ratio" (tiene que tener el nombre exacto)
- Se especifica el número de pines IN, OUT (ver esquemático y datasheet del componente).

Part Number	Component Type	Input Pin	Output Pin
14,17	N/A	N/A	N/A
000.957-IPF	IC Transceiver Bidirectional	16	4
000.982-IPF	IC Linear	8	2
000.990-PF	Sensor Temperature	7	1
001.944-IPF	IC QUAD Buffer	27	6
001.950-IPF	IC Digital. I2C Bus Translator	8	4
001.960-IPF	IC Transceiver Bidirectional	14	7
001.972-IPF	IC Digital. I2C Bus Translator	21	16
001.975-IPF	IC Digital ETC	5	4
001.977-IPF	IC Digital. I2C Bus Translator	8	6
005.210-PF	IC QUAD Buffer	10	4
005.212-PF	IC Differential Line Driver and Receiver	12	1

Compartir vista

Este equipo > Google Drive (G:) > .shortcut-targets-by-id > 1MC3gsk350qC6LiPP0KJevq6ZiOVtFvq > 02_INGENIERIA > 00_TEMPLATES > PLANTILLAS DOCUMENTACIÓN > FMECA_FMES > Templates

Nombre	Fecha de modificación	Tipo	Tamaño
old	15/06/2023 10:07	Carpeta de archivos	
1 - FMECA_Template_v3.1_1	15/06/2023 10:06	Hoja de cálculo habilita...	146 KB
2 - FMES_Template_v1.6	13/07/2020 9:59	Hoja de cálculo habilita...	101 KB
Revisiones para el FMECA_v1	05/11/2020 16:15	Documento de Microso...	16 KB
Z - FMECA_v1_0 - Instrucciones	15/06/2023 10:03	Documento de Microso...	24 KB

VELASCO

Hojas necesarias para la generación de la hoja
"FMEA" ("BOL2", "FM Ratio", "Refs")

Schematic Reference	Part Number	Definition	Schematic Sheet	Function	FMECA Item/Component	FMECA Description	FMECA Item Ref	FMECA Component Type	Analysis	Input Pins	Output Pins	FMECA Failure Rate
C1	011.809-PF	capacitor 1uf	10-FPGA_PWR	Power Management	Def: capacitor 1uf	tion: Power ManagementSchematic Sheet: 10-FPGA_P	10.10.1	Ceramic Chip Capacitor		N/A	N/A	0,033737294
C10	011.809-PF		13-FPGA_BANK_33_34_35	ARINC, RSxxx	P/N: 011.809-PF	on: ARINC, RSxxxSchematic Sheet: 13-FPGA_BANK_33_3	10.20.1	Ceramic Chip Capacitor		N/A	N/A	0,05466246
C100	011.830-PF		22-PEX8718_IO	PCI	P/N: 011.830-PF	Function: PCISchematic Sheet: 22-PEX8718_IO	10.30.1	Ceramic Chip Capacitor		N/A	N/A	0,028938034
C101	011.161-PF		10-FPGA_PWR	Power Management	P/N: 011.161-PF	tion: Power ManagementSchematic Sheet: 10-FPGA_P	10.10.2	Ceramic Chip Capacitor		N/A	N/A	0,054653935
C102	011.806-PF		14-FPGA_BANK_12_13_14	ARINC, RSxxx	P/N: 011.806-PF	on: ARINC, RSxxxSchematic Sheet: 14-FPGA_BANK_12_1	10.20.2	Ceramic Chip Capacitor		N/A	N/A	0,047510549
C103	011.806-PF		14-FPGA_BANK_12_13_14	ARINC, RSxxx	P/N: 011.806-PF	on: ARINC, RSxxxSchematic Sheet: 14-FPGA_BANK_12_1	10.20.3	Ceramic Chip Capacitor		N/A	N/A	0,047510549
C104	011.809-PF		14-FPGA_BANK_12_13_14	ARINC, RSxxx	P/N: 011.809-PF	on: ARINC, RSxxxSchematic Sheet: 14-FPGA_BANK_12_1	10.20.4	Ceramic Chip Capacitor		N/A	N/A	0,05466246
C105	011.806-PF		14-FPGA_BANK_12_13_14	ARINC, RSxxx	P/N: 011.806-PF	on: ARINC, RSxxxSchematic Sheet: 14-FPGA_BANK_12_1	10.20.5	Ceramic Chip Capacitor		N/A	N/A	0,047510549
C106	011.806-PF		14-FPGA_BANK_12_13_14	ARINC, RSxxx	P/N: 011.806-PF	on: ARINC, RSxxxSchematic Sheet: 14-FPGA_BANK_12_1	10.20.6	Ceramic Chip Capacitor		N/A	N/A	0,047510549
C107	011.758-PF		21-PEX8718_PWR	PCI	P/N: 011.758-PF	Function: PCISchematic Sheet: 21-PEX8718_PWR	10.30.2	Ceramic Chip Capacitor		N/A	N/A	0,026920338
C108	011.830-PF		22-PEX8718_IO	PCI	P/N: 011.830-PF	Function: PCISchematic Sheet: 22-PEX8718_IO	10.30.3	Ceramic Chip Capacitor		N/A	N/A	0,028938034
C109	011.800-PF		26-M1553_01_POWER	MIL-STD-1553	P/N: 011.800-PF	ion: MIL-STD-1553Schematic Sheet: 26-M1553_01_PO	10.40.1	Ceramic Chip Capacitor		N/A	N/A	0,071978098

<
>
FMECA
FM Effect
PartNum
BOL2
FM Ratio
Hoja1
Refs
I-O
+
:
◀
▶

Hoja "BOL2": esta hoja es la lista de materiales BOM de la tarjeta.
Se deben completar los campos en azul oscuro.

Hojas necesarias para la generación de la hoja
“FMEA” (“BOL2”, “FM Ratio”, “Refs”)

0	Type	FMD Type	Failure Mode	Ratio	PIN	
1	Resistor Chip	Resistor, Fixed, Film, Thin	Resistance Variance	57,27%	Each pin	FI
1	Resistor Chip	Resistor, Fixed, Film, Thin	Open Circuit	32,38%	Each pin	FI
1	Resistor Chip	Resistor, Fixed, Film, Thin	Short Circuit	10,35%	Each pin	FI
2	Ceramic Chip Capacitor	Capacitor, Fixed, Ceramic	Capacitance Variance	62,78%	Each pin	FI
2	Ceramic Chip Capacitor	Capacitor, Fixed, Ceramic	Short Circuit	31,24%	Each pin	FI
2	Ceramic Chip Capacitor	Capacitor, Fixed, Ceramic	Open Circuit	5,98%	Each pin	FI
3	Capacitor Electrolytic Tantalum Chip	Capacitor, Fixed, Electrolytic, Tantalum	Short Circuit	55,96%	Each pin	
3	Capacitor Electrolytic Tantalum Chip	Capacitor, Fixed, Electrolytic, Tantalum	Capacitance Variance	28,35%	Each pin	
3	Capacitor Electrolytic Tantalum Chip	Capacitor, Fixed, Electrolytic, Tantalum	Open Circuit	15,69%	Each pin	
4	Connector	Connector	Open	100,00%	Each pin	Ir
5	Inductor	Inductive Device, Coil, Electrical	Broken/Opened Inductor	50,00%	Each pin	
5	Inductor	Inductive Device, Coil, Electrical	Shorted Inductor	50,00%	Each pin	

>
FMECA
FM Effect
PartNum
BOL2
FM Ratio
Hoja1
Refs
I-O
+
:

Hoja “FM Ratio”: esta hoja es la lista de tipos de componentes, modos de fallo y frecuencia de ocurrencia extraída de la FMD.16.

Si en esta lista no hay un tipo “adecuado” para un determinado componente se debe buscar en la FMD-16 y añadir a esta lista tras renormalizar los ratios de ocurrencia (ver lista FMD-16).

Hojas necesarias para la generación de la hoja
"FMEA" ("BOL2", "FM Ratio", "Refs")

B	C	D	E	F	G
Card	Reference		Function / Sheet - AVIO2353	Reference	
AVIO2353	10		Power Management	10	
MFCC8558_AVIO2353	20		ARINC, RSxxx	20	
MFCC8558_VGP2870	30		PCI	30	
AVIO2355	40		MIL-STD-1553	40	
PSU1449	50		I2C	50	
SSU2138	60		CONFIGURATION JTAG	60	
BKP2146	70		MULTI GIGAbit Transceiver for PCIe	70	
BPB2148	80		RS232 / RS485	80	
COB2247	90		CLOCK	90	
VGP2870	100		Reset	100	
HOP	110		Time stamp	110	
			ARINC Controller	120	
			Connectors	130	
			GPIO	140	
			Mechanic	150	
			Multiple	160	

> FMECA FM Effect PartNum BOL2 FM Ratio Hoja1 Refs I-O

Hoja "Refs": a partir de estos Ids para cada tarjeta y para cada función se crean las referencias numéricas mencionadas en la hoja "FMEA".

Documentos

PDF: FLAGS_presentation + MOM

PDF: FLAGS_FMEA_RFQ_1_0_ANZEN

PDF: AVIO-2353A0_HDD_1_0_0

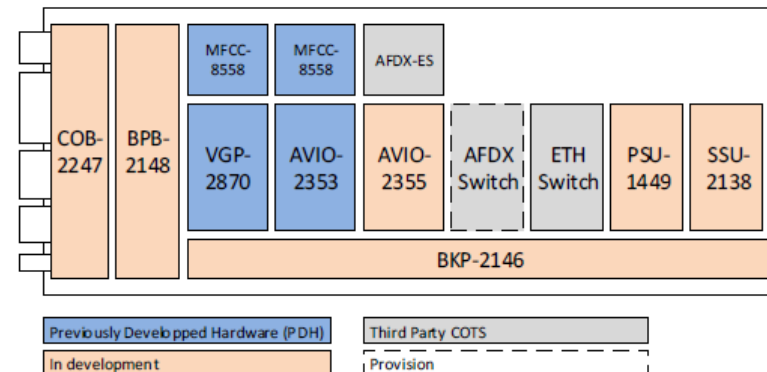
PDF: AVIO-2353BA43LNx_SE_3_0_0

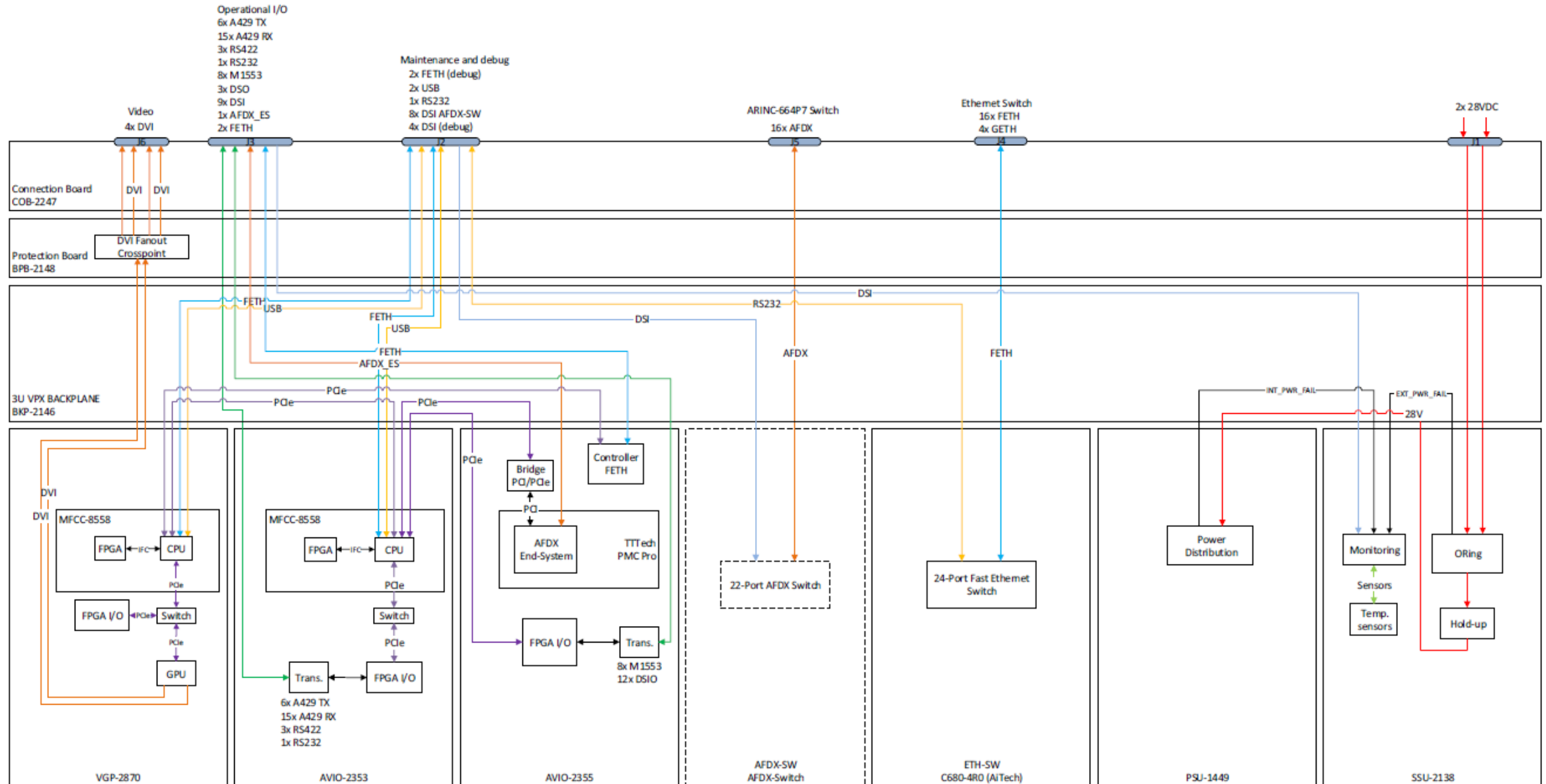
PDF: DATASHEET de componentes

Excel: FMEA

Resumen Hardware del Sistema. Ejemplo: Sistema FLAGS.

Item	Qty	Description	HW COTS
COB-2247	1	Connector Board	No
BPB-2148	1	Protection board against external aggression	No
BKP-2146	1	Backplane	No
AVIO-2353	1	Main processing unit carrier (3U OpenVPX form factor)	No
MFCC-8558	2	Main processing unit, Secondary processing unit (PMC form factor)	No
VGP-2870	1	Secondary processing unit carrier with video capability (3U OpenVPX form factor)	No
AVIO-2355	1	AFDX-ES carrier, Electronic for conditioning M1553 and DSIO signals (3U OpenVPX form factor)	No
SSU-2139	1	System Supervision Unit (3U OpenVPX form factor) with capacitor	No
PSU-1449	1	Power Supply Unit (3U OpenVPX form factor)	No
TTE-ES	1	AFDX End System (PMC/PCIE variant)	Yes (TTTech)
C680-R0	1	Ethernet Switch (3U OpenVPX form factor)	Yes (AiTech)

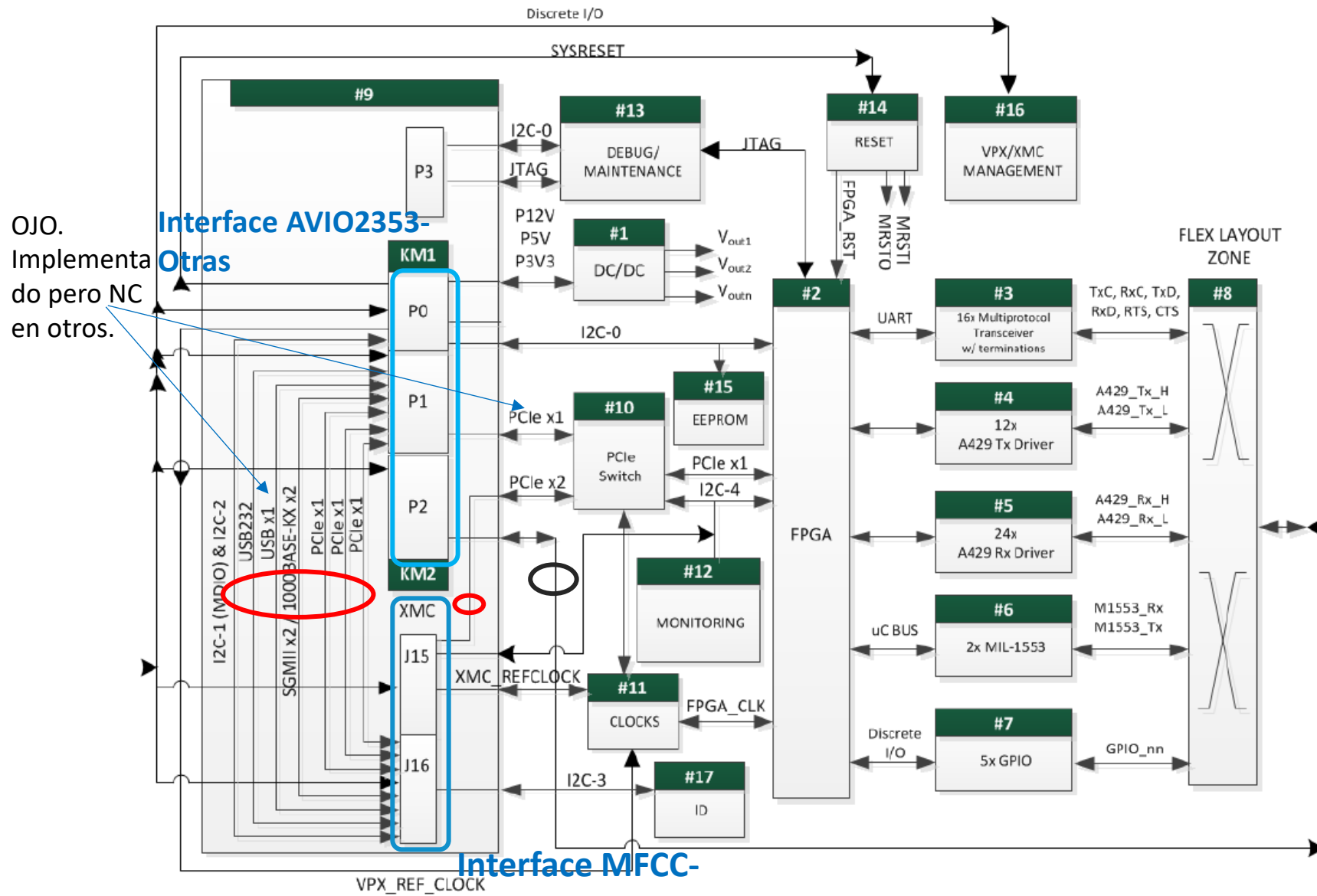
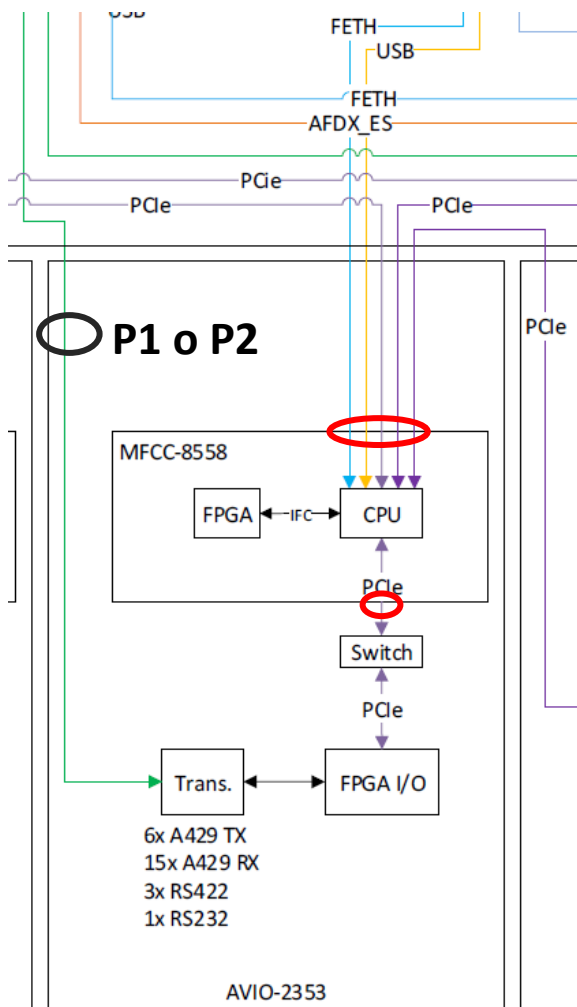




Resumen Interfaces Sistema FLAGS.

I/O Type	FLAGS Quantity	FLAGS Details
M1553	4	Dual MIL-STD-1553B bus interfaces are available on the system
A429_TX	6	ARINC-429 transmission interfaces are available on the system
A429_RX	15	ARINC429 reception interfaces are available on the system
RSL	3	RS-422 bus interfaces are available on the system
	1	RS-232 bus interface available on the system
	1	RS-232 used for maintenance of Ethernet switch
FETH	2	Operational ethernet interfaces are available on the system
AFDX-ES	1	AFDX end system is available on the system
DSO	3	DSO for operation: 1 general avionic purpose, 2 for internal health monitoring
DSI	9	DSI for operation: 2 general avionic purpose, 3 for installation, 2 for in-air/on-ground indication and 2 for GPU disabled
SRV_DSI	2	DSI to set maintenance mode
	2	DSI for FLAGS reset
	8	DSI for ETH switch configuration (5 used)
FETH-SW	16	FETH switch ports are available on the system
GETH-SW	4	GETH switch ports are available on the system
AFDX-SW	16	AFDX switch ports are in provision on the system
DVI	4	DVI outputs from two different graphic streams
SRV_FETH	2	Ethernet interfaces active in maintenance mode
SRV_USB	2	USB interfaces active in maintenance mode
Power	2	28VDC power supply

Ejemplo: Board AVIO2353 (portador MFCC).



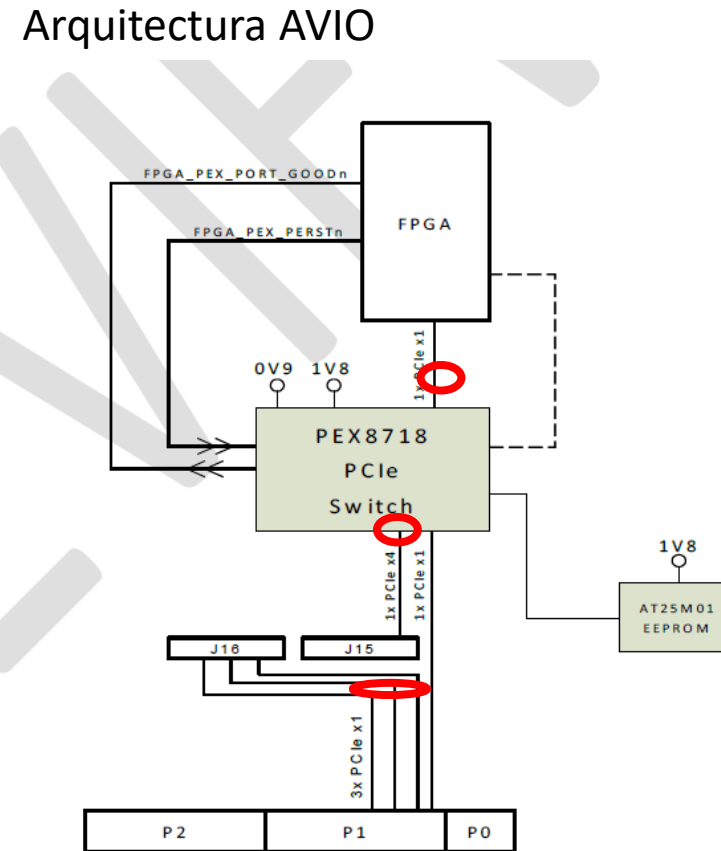
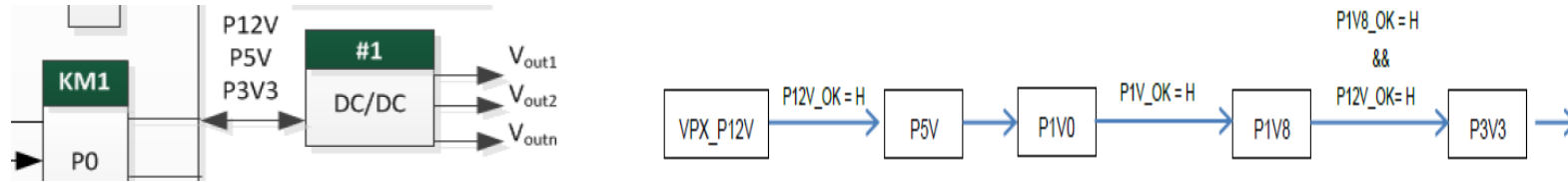


Figure 7 – PCIe

Bloque #1 DC/DC



Secuencia y monitorización de fuentes de alimentación: determinar cuando sea posible si se produce un fallo en “cadena” o un fallo aislado.

Loss of 12V power source.-> Loss of secondary power sources generation->Loss of AVIO2353 operations.
Mitigation/Remark: Monitoring.

Cortocircuito Condensadores de desacoplo: consideramos que nos lleva a “Pérdida” de la línea de alimentación asociada.

En este caso, dado que no tenemos conocimiento del sistema más allá de FLAGS, nuestros efectos a nivel “System” (L1) serán efectos referidos a las interfaces (“canales”) del sistema FLAGS:

- “Loss of A429_TX channel”
- “Erroneous A429_TX channel”
- “Degraded A429_TX channel”
- Etc.

Si se conoce que algún canal hace “interface” con otra tarjeta se debe indicar (entre parentesis)

- Loss of PCIe_01_Tx (AVIO2353/VGP2870)

Otros efectos a nivel “System” habituales son:

- “Loss of AVIOxx operations”
- “Erroneous AVIOxx operations”
- Etc.